

Parent Function Families

Name: _____

Pre-Calculus | Guided notes — we fill these in together

Vocabulary

Parent function: the _____ form of a family — no shifts, stretches, or reflections.

Domain / Range: domain is the set of allowed _____-values; range is the set of resulting _____-values.

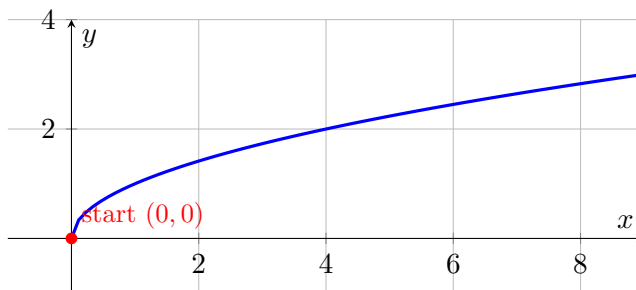
We do together. Name the parent of $f(x) = \sqrt{x}$ and give its domain.

Step 1. Match the rule to a known family: this is the _____ parent.

Step 2. Recall its shape: it starts at the point _____ and curves up to the right.

Step 3. A square root needs a non-negative inside, so the domain is _____.

See it:



$y = \sqrt{x}$ — starts at the origin, domain $x \geq 0$.

You Try. Solve each. Show every step, then name the answer.

a) Name the parent of $f(x) = \frac{1}{x}$ and give its domain.

b) Name the parent of $f(x) = |x|$ and give its range.

Summary (fill in): To name a parent: match the _____ to a family, recall its _____, then state the _____.

Transformations of Functions

Name: _____

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Vocabulary

Translation: a _____ of the graph — $f(x)+k$ moves it _____, and $f(x-h)$ moves it _____.

Reflection: a flip: $-f(x)$ flips over the _____, and $f(-x)$ flips over the _____.

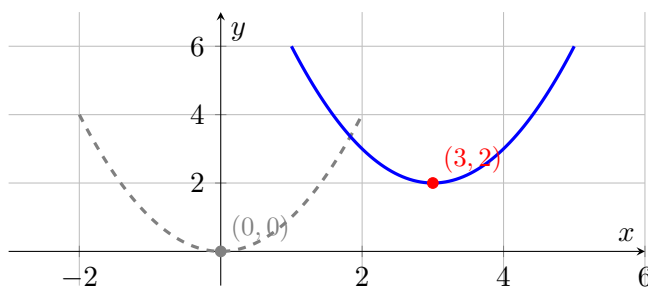
We do together. Describe how $g(x) = (x - 3)^2 + 2$ transforms the parent x^2 .

Step 1. Inside the square, $x - 3$ shifts the graph _____ (opposite the sign).

Step 2. The $+2$ outside shifts it _____.

Step 3. So the vertex moves from $(0, 0)$ to _____.

See it:



parent (dashed) $\rightarrow$ right 3, up 2 (solid).

You Try. Solve each. Show every step, then name the answer.

a) Describe $h(x) = -x^2$ from the parent.

b) Give the vertex of $y = (x + 1)^2 - 4$.

Summary (fill in): $f(x - h) + k$ shifts a graph _____; a leading $-f(x)$ _____ it over the x -axis.

Domain & Range

Name: _____

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Vocabulary

Domain: all the _____ values (x) a function is allowed to use.

Range: all the _____ values (y) the function actually produces.

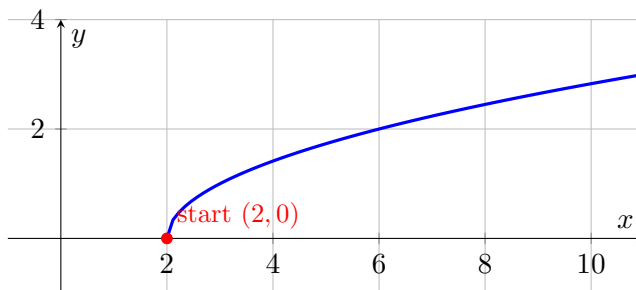
We do together. Find the domain and range of $f(x) = \sqrt{x-2}$.

Step 1. A square root can't hold a negative: set $x - 2 \geq 0$, so the domain is _____.

Step 2. The smallest output is at $x = 2$, where $f(2) = \underline{\hspace{1cm}}$, then it grows.

Step 3. So the range is _____.

See it:



$y = \sqrt{x-2}$ — domain $x \geq 2$, range $y \geq 0$.

You Try. Solve each. Show every step, then name the answer.

a) Give the domain of $f(x) = \frac{1}{x-5}$.

b) Give the range of $f(x) = x^2 + 1$.

Summary (fill in): Watch two traps: a _____ can't be 0, and a _____ can't hold a negative.

Maximum & Minimum

Name: _____

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Vocabulary

Max / min: the _____ of a parabola (the vertex); opens up gives a _____, opens down gives a _____.

Increasing / decreasing: read left to right — the graph goes _____ (increasing) or _____ (decreasing).

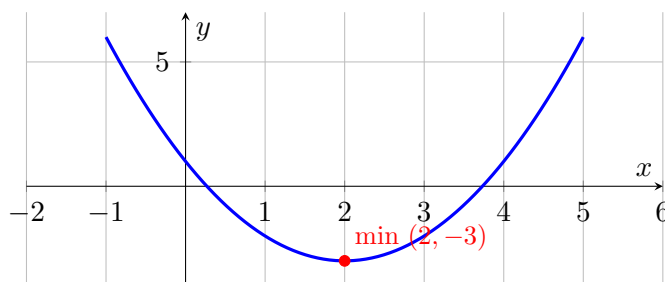
We do together. For $y = (x - 2)^2 - 3$, give the vertex, state max or min, and the intervals.

Step 1. Vertex form $a(x - h)^2 + k$ gives vertex _____.

Step 2. Leading coefficient is +1 (opens up), so the vertex is a _____.

Step 3. It decreases for _____ and increases for _____.

See it:



opens up $\Rightarrow$ minimum at $(2, -3)$.

You Try. Solve each. Show every step, then name the answer.

a) Give the vertex and max/min of $y = -x^2 + 4$.

b) Give the intervals of $y = x^2$.

Summary (fill in): Opens _____ $\Rightarrow$ minimum; opens _____ $\Rightarrow$ maximum; the graph turns at the _____.

Symmetry: Even & Odd Functions

Name: _____

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Even function: symmetric across the _____; algebraically $f(-x) = \underline{\hspace{2cm}}$.

Odd function: symmetric about the _____; algebraically $f(-x) = \underline{\hspace{2cm}}$.

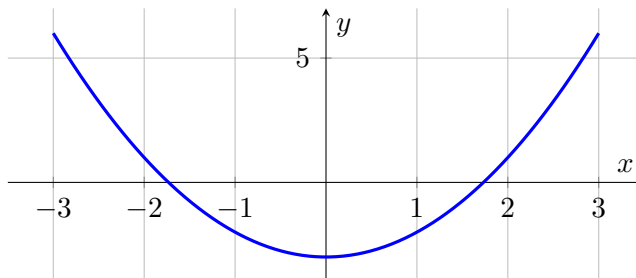
We do together. Is $f(x) = x^2 - 3$ even, odd, or neither?

Step 1. Replace x with $-x$: $f(-x) = (-x)^2 - 3 = \{\{x^2 - 3 - 2cm\}\}$.

Step 2. Compare to $f(x)$: they are _____, so $f(-x) = f(x)$.

Step 3. Therefore f is _____ (symmetric across the y -axis).

See it (even = y-axis mirror):



$y = x^2 - 3$ — the y -axis is a mirror line (even).

You Try. Solve each. Show every step, then name the answer.

a) Is $f(x) = x^3$ even, odd, or neither?

b) Is $f(x) = x^2 + x$ even, odd, or neither?

Summary (fill in): Compute $f(-x)$: equals $f(x) \Rightarrow$ _____; equals $-f(x) \Rightarrow$ _____; otherwise _____.